

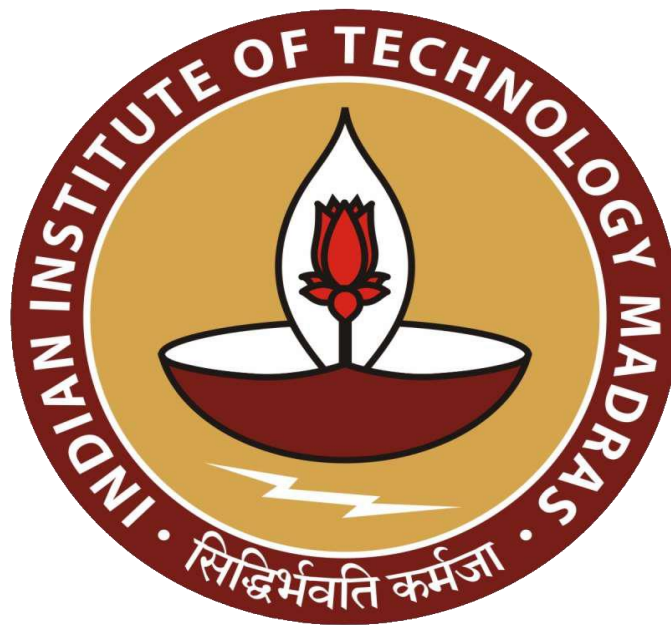
Data-Driven Optimization in Bottle Manufacturing

A Mid-Term report for the BDM capstone Project

Submitted by

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1. Executive Summary

This report is based on data collected from SP Enterprises, a small plastic bottle manufacturing unit in Bhagwanpur, Haridwar, Uttarakhand. The raw material procurement and client order data were digitized, **cleaned** for missing values, and standardized to ensure consistency. The final dataset includes two sheets: one with raw material purchase details (date, type, quantity, and price per kg), and another with client order details (order date, client name, size of bottles and number of bottles ordered).

Descriptive statistics were used to summarize the data and identify key patterns. For raw materials, pivot tables helped calculate total quantities and costs over time, highlight commonly used materials, and check price trends. HDPE and PET were the most frequently used materials, with HDPE showing the highest contribution to both volume and cost. The client order data was analyzed to understand order volumes, popular bottle sizes, and demand trends. This step provided a clear picture of procurement behavior and customer order patterns, critical for planning and forecasting. Aligning procurement with actual order trends can reduce extra stock and save money. It also helps in making sure enough material is available when needed.

From January to May 2025, the **combination chart(line and bar)** was created. It showed some mismatches between procurement and bottle demand. In March, raw material procurement peaked while bottle orders dropped, suggesting possible overstocking. HDPE, PET, and PP were the most procured materials overall. The **stacked column chart** was created to observe bottle demand. 100 ml bottles were highly demanded across all months, with growing demand for 5000 ml bottles in April and May. May showed the highest combined procurement and bottle orders, reflecting increased production activity. These gaps highlight the need for better planning and forecasting. **Conditional Formatting** highlights a fluctuating trend in monthly purchases across different materials.

This analysis supports growth of manufacturing unit while addressing some problems in its operation. It helps the company understand when and how much material to purchase based on actual demand. These improvements can lead to smoother operations, better customer satisfaction, and higher profits over time.

2. Proof of Originality

2.1 Details:

- Company name: S P Enterprises
- GST No: 05AETFS4444M1ZC
- Type of organization: B2B (Business-to-Business)
- Location: Makhanpur, Bhagwanpur, Haridwar, -247661
- Owner name: Pankaj Kamboj
- Owner's Phone No: 9758859352
- Number of Employees: 3
- Working Time: 09:00AM -06:00PM



Image 2.1



Image 2.2

2.2 Proof Link:

- Images, Video, Authorization Letter

<https://drive.google.com/drive/folders/1dzrmlIwwWbMDKXQ1qezs7rw5abXS65DfK?usp=sharing>

3. Meta Data

- **Source of Data:** The data is mainly collected from handwritten records.
- **Format:** Initially in raw format, the data was later digitized and structured in Microsoft Excel.
- **Time Period Covered:** Approximately 5-6 months of data has been compiled and maintained in the Excel file.
- **Data Entry:** Manual entry was involved during the digitization process.

Clean dataset link: https://docs.google.com/spreadsheets/d/1_yip7U_9leNoJXzI6f6JzH6RPH-K10v7/edit?usp=drive_link&ouid=101131464344423523653&rtpof=true&sd=true

Data collection dates: 4-Jan-2025 to 31-May-2025

Data dimensions:

- Raw Material Data: The dataset contains **6 columns** and **102 rows**.

Column Name	Description	Data Type
Date	Dates in which raw material was purchased	Date
Material	Name of Material purchased	Text
Supplier	Name of company who supplied raw material	Text
Quantity_kg	Quantity of raw material purchased	Number
Price_per_kg	Price of 1 kg of raw material	Number
Total Price	Total Price of Quantity purchased	Number

Table 3.1

- Client Order Data: The dataset contains **4 columns** and **297 rows**.

Column Name	Description	Type
Date	Date on which order was placed	Date
Client	Name of company who gave order	Text
Bottle_Size_ml	The size of bottles which were ordered	Number
No_of_Bottles	Number of bottles were ordered	Number

Table 3.2

4. Descriptive Statistics

4.1 Raw material analysis:

	Mean	Median	Mode	Standard Deviation	Standard Error	Sample Variance	Range	Maximum	Minimum	Sum	Count
HDPE	232.857	220	200	37.307	8.141	1461.43	150	300	150	4890	21
HIPS	203.684	200	200	13.82	3.172	191.228	70	250	180	3870	19
PP	185.909	200	200	38.378	8.182	1472.94	200	250	50	4090	22
PET	217.619	200	200	24.862	5.425	649.048	70	250	180	4570	21
PVC	144.736	150	150	30.798	7.065	948.538	100	200	100	2750	19

Table 4.1 Raw Material Statistics

- This analysis explains how different raw materials were purchased. The **mean** purchase quantity was highest for **HDPE** at 232.86 kg, followed by PET at 217.62 kg. **PVC** had the lowest mean of 144.74 kg, showing it was bought in smaller amounts compared to the others. The **median** and **mode** for most materials were **200 kg**, meaning 200 kg was a common and regular order size.
- To understand how much the quantities varied, I look at the **standard deviation**. **PP** and **HDPE** had the highest standard deviations (around **38 kg** and **37 kg**), showing their purchase quantities changed a lot. On the other hand, HIPS had a low standard deviation of 13.82 kg which means it was bought in more **consistent amounts**.
- We also looked at the **sample variance** to measure how much the data is spread out. **PP** and **HDPE** again had high variances (1472.94 and 1461.43), while **HIPS** had the lowest variance (191.23), which confirms that HIPS purchases were the most stable. This means that when the company bought HIPS, the quantities stayed close to the average every time unlike **PP** or **HDPE**, where the purchase amounts fluctuated more.
- The **total quantity** of **HDPE** purchased was **4890 kg**, the highest among all, followed by PET at 4570 kg. **PVC** had the lowest total at 2750 kg, showing it was the least used. The number of times the material was purchased was between **19 and 22** for each material, which is enough to understand the purchase trend.

- From this data, we can say that **HDPE and PET** are the main raw materials, used more frequently and in larger quantities. **HIPS** appears to be a regularly purchased material in steady amounts, which helps in **inventory planning**. **PP**, even though used significantly, had very inconsistent purchase amounts, suggesting it may be bought only during urgent or unexpected needs.
- These insights point to an **inefficient raw material procurement strategy**, where some materials are ordered unpredictably and without a clear pattern. The high variability in HDPE and PP purchases results in either **stock shortages** or **excess inventory**, both of which increase costs and affect production continuity. A lack of consistent procurement planning leads to supply mismatches, rushed orders, and missed opportunities for supplier negotiation. By identifying which materials are stable and which ones are not, the company can revise its strategy to improve **demand forecasting, and overall procurement efficiency**.

4.2 Client Order Analysis:

	AVSR	Ayurgram	Farmatech	Ishwer India	Js vet	Rasko India	Siddeshwari	Skv vet
Mean	1031.25	977.631	818.292	1085.294	960.294	870	917.948	974.359
Median	800	825	700	975	800	750	700	800
Mode	800	800	700	500	800	550	700	750
Standard Deviation	437.505	402.81	410.298	637.529	430.253	373.373	499.01	379.381
Standard Error	77.34	65.344	64.077	109.335	73.787	59.01	79.9	60.749
Sample Variance	191411.3	162256.4	168344.5	406443.9	185118.1	139333.3	249011.5	143930.5
Range	1800	1600	2000	2200	1450	1950	1800	1600
Maximum	2350	2000	2400	2500	1950	2450	2200	2000
Minimum	550	400	400	300	500	500	400	400
Sum	33000	37150	33550	36900	32650	34800	35800	38000
Count	32	38	41	34	34	40	39	39

Table 4.2 Client Order Statistics

- This analysis is based on bottle order data from different clients. Among all, **Ishwer India** placed the highest average number of orders, with a mean of 1085 bottles, followed by AVSR and Js Vet. The lowest average orders came from **Farmatech (818 bottles)**, indicating smaller or less frequent demand from that client. The **median** and **mode** values for most clients were around **700 to 800 bottles**, suggesting these were the most common order sizes.
- The **standard deviation** and **sample variance** help us understand how much each client's order quantities changed. **Ishwer India** had the highest **standard deviation** (about **638**), meaning their orders varied a lot sometimes very high, sometimes very low. This kind of variation leads to **production bottlenecks**, where sudden large orders may be hard to fulfill quickly, or small orders may lead to underutilized capacity. On the other hand, **Rasko India** and **Skv Vet** had more **stable order patterns**, which are easier to plan for.
- Clients like **Siddeshwari** also showed high variance in orders, which could signal **unpredictable demand**. This kind of inconsistency makes it hard for the manufacturing unit to maintain the right amount of raw material and manpower, leading to either **overproduction or shortages**.
- Looking at the **total orders** placed by each client, **Skv Vet** had the highest at **38,000 bottles**, followed closely by Ayurgram and Siddeshwari. This indicates that even if their average order size isn't the highest, they have been consistent in placing large numbers of total orders. The **count** of orders also shows frequency for example, **Farmatech** had **41** orders which is the most, even though their average was low, meaning their demand was small but regular.
- Understanding these trends helps the company **forecast production more effectively**. Clients with consistent orders can be handled through regular scheduling, while clients with high variation may require **buffer stock**, **flexible labor**, or **early communication** to avoid last-minute issues. Such analysis can also help in **client classification**, identifying stable clients for bulk production and flagging unpredictable ones that may need custom handling or separate planning.
- Overall, this descriptive analysis not only shows order quantities but also highlights risks and opportunities in the client relationship and production strategy.

5. Method of Analysis with Justification

5.1 Data Cleaning and Consolidation:

- **Process:** The data was mainly collected from handwritten notes. It was then entered into Excel to make it easier to work with. While entering, some spelling mistakes and duplicate entries were found and corrected. Dates were also **cleaned** and changed into the same format. After cleaning, all the data was **combined** into one Excel sheet so that it could be used for analysis easily.
- **Justification:** Since the data came from different sources and was not in proper format, cleaning was important to avoid mistakes. Putting all the data together in one place helped in understanding and comparing it better. For a small company like SP Enterprises, this dataset is essential to understand **raw material purchase trend** and **order from client**. Without this step, wrong conclusions could have been drawn from incorrect totals or averages.

5.2 Descriptive Statistics Analysis:

- **Process:** Once the data was cleaned and properly arranged, **descriptive statistics** were used to understand the overall patterns in the data. This type of analysis helps to **summarize large sets of data** in a simple and understandable way. Some statistical measures such as **mean, median, mode, standard deviation, sample variance, standard error, sum, maximum, minimum** were calculated for different categories. These included the number of bottles ordered by each client, the monthly demand, and the quantities of raw materials purchased. For example, I used **Excel formulas** like =SUM(), =AVERAGE(), =MAX(), and =MIN() to find out how many bottles were ordered in total, which client ordered the most, and which month had the highest or lowest demand. I also used **pivot tables** to group the data and see totals for each bottle size or material type. This helped us understand variation in number of bottles ordered by different clients and which raw materials were used more often.
- **Justification:** Using descriptive statistics is a simple and effective way to get a clear overview of the data without needing any complex tools. It helps in **identifying trends**,

making comparisons, and finding key insights. This step was very important before doing any advanced analysis because it showed the basic structure and variation in the data. In this project, descriptive statistics helped to highlight the most important areas such as top clients, preferred bottle sizes, and the frequency of raw material purchases, which are useful for decision-making and planning. By calculating measures like mean, median, mode, and standard deviation, we were able to understand which materials and clients showed stable behavior and which ones had unpredictable trends. This statistical summary acted as the foundation for identifying **inefficient raw material procurement strategy, demand-supply imbalances and production bottlenecks**, aligning directly with the problem statement.

5.3 Correlation Analysis:

- **Process:** To understand how raw material purchases relate to bottle demand and price, correlation analysis was done. First, monthly totals were calculated for bottle orders, raw material quantities, and prices. Then, Excel's Correl() function was used to find two relationships:

- (1) between raw material quantity and bottle demand,
- (2) between raw material price and quantity purchased.

The result showed a **–0.3 correlation between material and demand**, and **–0.54 between price and quantity**.

- **Justification:** This analysis was used to check whether procurement decisions are properly linked with actual production needs and whether the company is **making price-based buying decisions**. The weak correlation between material and demand shows that raw material is not always purchased based on bottle orders. This leads to **storage issues, or stock shortages**, especially if the material arrives too early or too late. The negative correlation between price and quantity suggests that **purchases increase when prices are low**, which is a **cost-saving strategy**. However, without tracking actual demand, this may also result in **overstocking**. So further analysis like **ABC Segmentation, Forecasting** techniques like **Linear Regression, Moving Averages** will be done to make better decision for purchasing raw material and predicting demand from clients.

6. Results

6.1 The raw material procurement data from January to May 2025 highlights a **fluctuating trend** in monthly purchases across different materials. Overall, the total quantity of materials procured increased over the months, with a noticeable peak in **May**. Among the five materials, **HDPE** emerged as the most heavily procured, followed by PET and PP reflecting their crucial role in production. The **conditional formatting (color coding)** indicates monthly highs and lows, helping identify anomalies and procurement spikes.

Row Labels	Jan	Feb	Mar	Apr	May	Grand Total
HDPE	1050	850	1170	680	1140	4890
HIPS	380	820	850	800	1020	3870
PET	840	870	900	950	1010	4570
PP	760	830	780	750	970	4090
PVC	430	550	640	610	520	2750
Grand Total	3460	3920	4340	3790	4660	20170

Figure 6.1 Conditional Formatting

Key observations:

- HDPE had the highest overall procurement (4890 kg), with significant peaks in March (1170 kg) and May (1140 kg), but a noticeable drop in April (680 kg).
- PET showed consistent procurement each month, indicating stable demand and usage.
- HIPS started low in January (380 kg) but steadily increased to 1020 kg in May, suggesting growing reliance or demand.
- PP purchases followed a similar upward trend, reaching their highest in May (970 kg).
- PVC had the lowest overall procurement (2750 kg) and fluctuated irregularly, with no strong upward or downward trend.
- April was a relatively low procurement month for most materials, possibly due to stock utilization or decreased production needs.

6.2 The stacked column chart illustrates the monthly demand for bottles of different sizes from January to May 2025. Each bar represents the total number of bottles demanded in that month, broken down by bottle size 100ml to 5000ml. The chart gives insights into **size-wise demand trends** and **overall bottle requirements** over time.

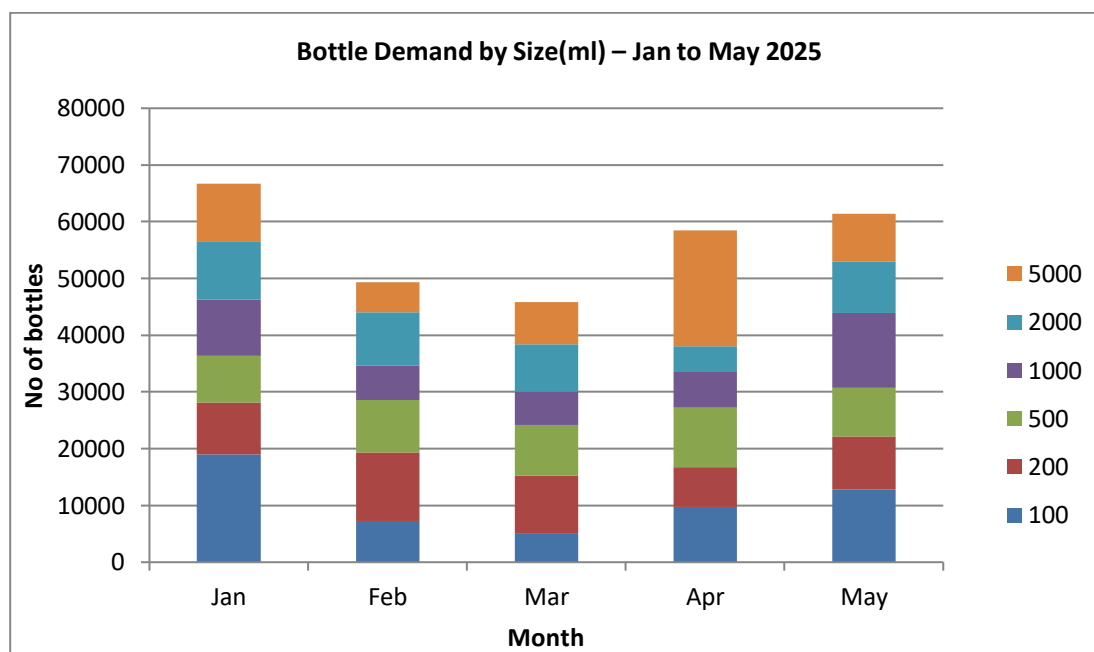


Figure 6.2 Stacked Column Chart

Key Observations:

- January had the highest total bottle demand, reaching over 65,000 bottles, indicating a strong start to the year.
- March saw the lowest demand, less than 50,000 bottles, showing a temporary dip in bottle requirements.
- Bottle demand recovered in April and May, with a gradual increase possibly due to seasonal trends or marketing pushes.
- 100 ml bottles consistently dominate demand across all months, forming the largest share of each month's total.
- 500 ml and 1000 ml bottles also show steady demand and contribute significantly to total volumes.

6.3 This **combination chart** compares **raw material procurement (bar chart)** with the **number of bottles ordered (line graph)** from January to May 2025. It helps identify alignment or mismatch between supply and demand trends over time.

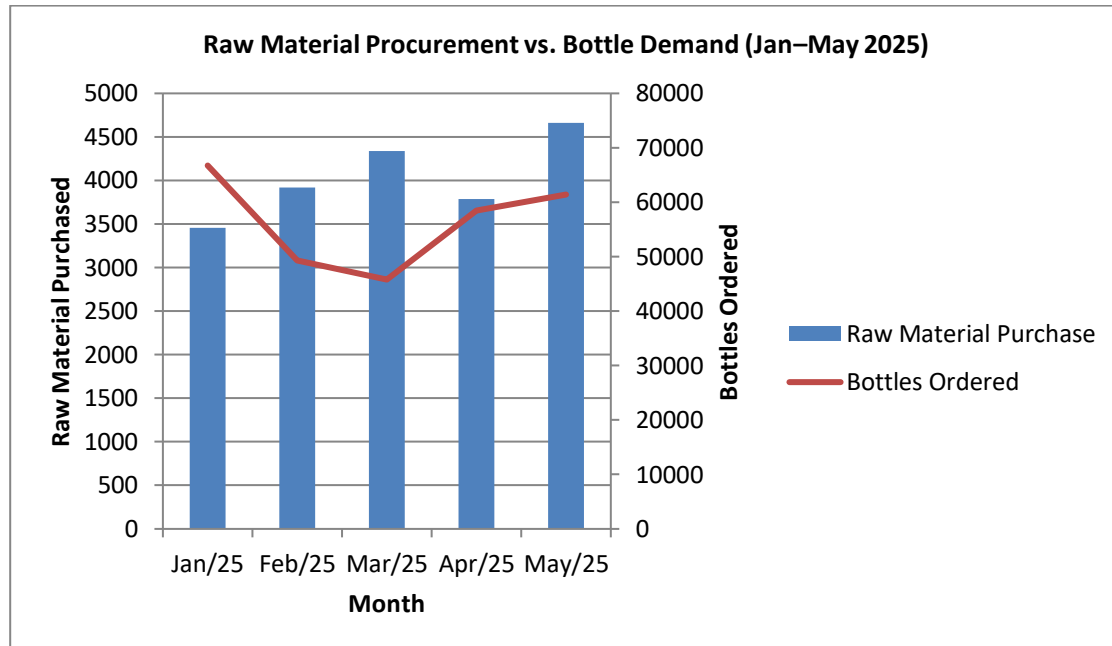


Figure 6.3 Combo Chart

Key Observations:

- January 2025 shows lower raw material purchase (~3460 kg) but very high bottle demand (~66,750 bottles), indicating reliance on existing inventory or underprocurement.
- In March 2025, raw material procurement peaked (~4340 kg) while bottle orders dipped (~45,800 bottles), suggesting possible overstocking or incorrect demand forecasting.
- April shows a recovery in bottle demand and moderate procurement, possibly balancing past inventory adjustments.
- May 2025 recorded the highest raw material purchase (~4660 kg) and high bottle orders (~61,450), which may indicate confirmed future demand.